



UNIVERSITY OF EDINBURGH
Business School

2025- 26

Predictive Analytics for Business
Individual Essay

B240834

Word count: 3484

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February 27, 2026

1 Question 1

All summary statistics tables referenced in this section are provided in Appendix A.1.

As shown in Table 12, the oj.csv dataset (OJ) contains 1,728 observations across three categorical variables (Store, Feature and Brand) and three continuous variables (Price, Week and Logmove). The data spans 120 weeks and is balanced across the three brands (576 observations each) but is slightly unbalanced across stores (330-354).

In the aggregate pricing data (Table 12), prices range from \$0.94 to \$3.87 with a right skew (mean: \$2.30 > median: \$2.17), driven largely by Tropicana's premium position (mean: \$2.89) versus Dominick's budget strategy (mean: \$1.75). Brand appears to be the primary determinant of price; however, variation between stores exists, with Store 12 being the most expensive (mean: \$2.41), while Store 8 is the most affordable (mean: \$2.18).

Logmove (the logarithm of units sold) exhibits a mild right skew, with the aggregated values ranging from 6.356 to 12.724 (Table 12). The mean Logmove remains consistent across all three brands, ranging from 9.236 to 9.339 (Table 14), suggesting that Tropicana's brand equity effectively offsets its higher price point, allowing it to maintain sales volumes comparable to the budget-friendly Dominick's. Table 13 shows store-specific differences in Logmove; store 8 and 12 exhibit the highest mean Logmove (9.571 and 9.516, respectively), while store 2 exhibits the lowest (9.131). The maximum Logmove at each store greatly exceeds their means, indicating periods of substantially increased sales.

Feature advertisements are infrequent, occurring in only 23.55% of observations. Table 13 highlights consistent feature incident rate across stores (23.3%-23.7%), suggesting coordinated promotional campaigns. Minute Maid has the most promotionally aggressive strategy (165 features), whereas Tropicana relies more on brand equity (94 features) to achieve comparable sales volumes at premium prices.

2 Question 2

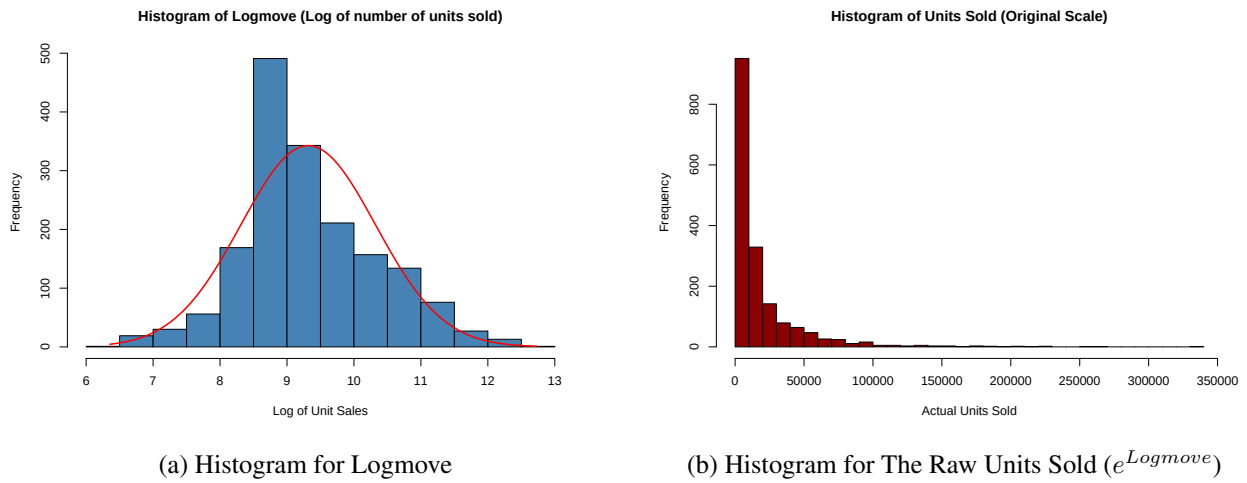


Figure 1: Demand Analysis of Orange Juice: Assessing sales volatility across log-transformed and original unit scales.

Figure 1a shows the distribution of Logmove and consists of 12 bins to balance excessive smoothing from too few bins with the increased noise from too many (Sturges' rule[Scott, 2009]). The auxiliary histogram illustrates the raw sales data ($e^{Logmove}$). Here, the number of bins was arbitrarily set at 30 to provide sufficient granularity for the distribution. In Figure 1a, the red line shows a normal distribution fitted to the data.

Figure 1a presents an approximately symmetric, unimodal histogram with a modal value in the interval spanning 8.5–9.0, with most observations concentrated between the values of 8 and 11. This distribution supports the finding in Section 1 that Logmove is mildly right-skewed due to its pronounced right tail; the lack of isolated bars in the histograms' tails suggests that the data doesn't contain any extreme or isolated outliers for Logmove. By comparing Figure 1a with Figure 1b, it is evident that by compressing extreme values, the log-transformation has helped the heavily skewed sales data distribution becomes more symmetrical and more consistent with normality. Thus, improving the variables' suitability for regression by stabilising variance.

In a business context, Figure 1a reveals that most store-brand-week combinations sell between approximately 3,000 (e^8) and 60,000 (e^{11}) units per week, with the most common sales volume being between 5,000–8,000 units. The extreme right skew of this data indicates that while baseline sales are consistent, there are periods of dramatically increased sales.

3 Question 3

Figure 2 details information on Logmove for each brand exclusively at Store 2. A boxplot was chosen, as it summarises the distribution of a continuous variable (Logmove) across categories (Brands), showing the central tendency, spread and outliers. Table 15 (Appendix A.2) quantifies these results and includes the mean and standard deviation to measure the volatility of brand-specific sales.



Figure 2: Logmove by Brand for Store 2.

Tropicana ($IQR = 0.932$, $\sigma = 0.646$) has the lowest interquartile range and standard deviation, indicating that its sales are more consistent and less volatile than Dominick's ($IQR = 1.181$, $\sigma = 1.071$) and Minute Maid ($IQR = 1.198$, $\sigma = 0.848$). The greater spread observed for Dominick's and Minute Maid is consistent with their more frequent use of feature advertisements.

Statistical outliers are classified using Tukey's fences ($k = 1.5$) [Dümbgen and Riedwyl, 2007]. Graphically, they appear as data points outside the bounds of the whiskers (range of Tukey's fences). Three outliers have been identified for Dominick's and one for Minute Maid, representing periods of unusually low or high sales. These outliers disproportionately inflate the standard deviation, as it is calculated by squaring deviations from the mean.

The high outliers could be explained by promotional activities that temporarily increase demand. The low-outlier seen in Dominick's data may reflect a period in which a competing brand reduced prices to challenge Dominick's price position, a temporary stock-out, or other adverse event.

All three brands exhibit similar central tendencies, with median Logmove values ranging from 8.891 (Minute Maid) to 9.122 (Dominick's), indicating that their typical weekly sales are comparable. The median line positions within the box (representing *IQR*) indicate that the distribution of Dominick's Logmove is relatively symmetric (central positioning), and that Minute Maid and Tropicana have right-skewed distributions, as their median lines sit closer to the lower quartile. These observations are supported by the auxiliary histograms, shown in Figure 3, which illustrate each brand's Logmove distribution for store 2.

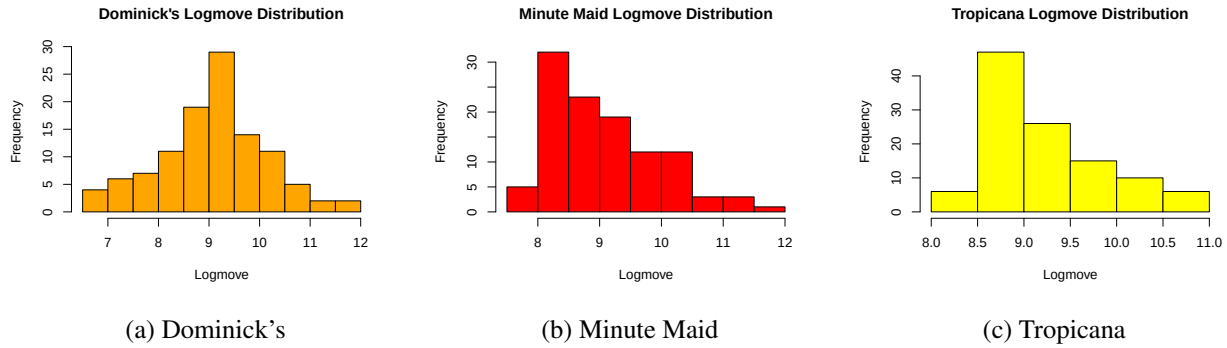


Figure 3: Histograms of Logmove for Store 2 Brands, illustrating varying degrees of skewness.

4 Question 4

4.1 Model Specification

To investigate the relationship between Logmove and the natural logarithm of Price, Logprice, a simple linear regression model was created.

$$\text{Logmove} = \beta_0 + \beta_1 \text{Logprice} + \epsilon \quad (1)$$

Logprice is used here instead of price, since Logmove is also logarithmic; the coefficient β_1 represents price elasticity i.e percentage change in units sold per 1% change in price. The intercept (β_0) represents the expected value of Logmove when Logprice is zero. The error term (ϵ) represents the part of Logmove that Logprice alone cannot explain.

4.2 OLS estimator

To estimate the parameters, Ordinary Least Squares (OLS) was used. OLS minimises the residual sum of squares and produces the Best Linear Unbiased Estimator (BLUE) under the Gauss-Markov assumptions of linearity, homoscedasticity, no (or little) multicollinearity and the independence of residuals [Howland and Barreto, 2005].

Additionally, the normality of residuals is also required for valid statistical inference. Model 1 contains one predictor, trivially satisfying the multicollinearity assumption.

4.3 Validation of the OLS assumptions

All hypothesis tests in this report are conducted at the 5% significance level.

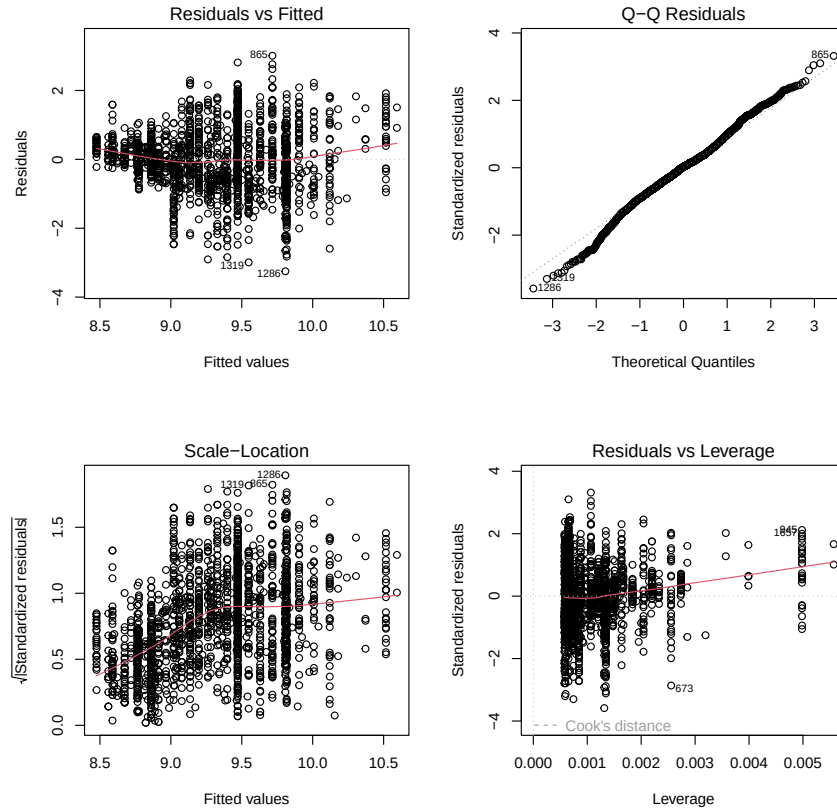


Figure 4: OLS Regression Diagnostic Plots: Residuals vs Fitted, Normal Q-Q, Scale-Location, and Residuals vs Leverage.

4.3.1 Assumption 1 - Linearity

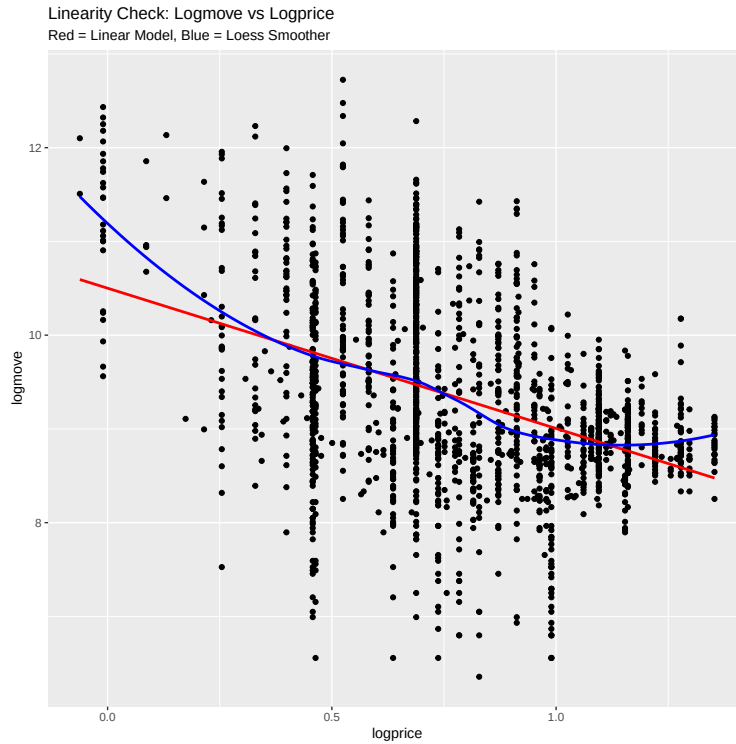


Figure 5: log(Price) vs Logmove: Investigating Linearity

The LOESS line in Figure 5 deviates from the regression line in a U-shape, consistent with the curvature in the Residuals vs fitted plot, both of which indicate a non-linear relationship between Logmove and Logprice. A curvature test ($t = 5.4104$, $p = 7.170 \times 10^{-8}$), detailed in Table 1, evaluated whether including Logprice^2 improved the model's fit, thereby testing the linearity of the relationship between Logprice and Logmove. The test rejected H_0 of correct linear specification, confirming that the linearity assumption is violated, which suggests that the model's estimates may be biased.

Table 1: Curvature test for Logprice Hypotheses and Results

Hypotheses			
H_0 : The model is correctly specified (linear relationship between Logprice and Logmove).			
H_1 : The model is incorrectly specified (non-linear relationship between Logprice and Logmove).			
Variable	Test Statistic (t)	p-value	Significance
<i>LogPrice</i>	5.4104	7.170×10^{-8}	***

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

4.3.2 Assumption 2 - Normality of the Residuals

Given the large sample size of OJ ($n = 1728$), a Shapiro-Wilk test was not performed, as at this scale the test becomes too sensitive, rejecting H_0 of normality for deviations that are detectable but practically inconsequential. Instead, this assumption is diagnosed visually using the Q-Q Residuals plot (Figure 4), which shows the residuals align closely with the normal reference line, satisfying assumption 2.

4.3.3 Assumption 3 - Homoscedasticity

The Scale-Location diagnostic plot seen in Figure 4 shows the spread of residuals increasing systematically with the fitted values. Indicating heteroscedasticity, a violation of Assumption 3, which makes the standard errors unreliable.

4.3.4 Assumption 4 - Independence of the residual error terms

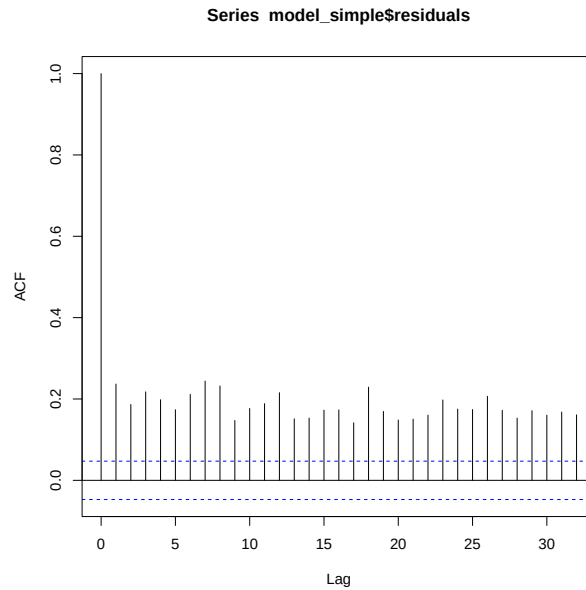


Figure 6: Autocorrelation Function (ACF) Plot of Residuals.

In the ACF plot (Figure 6), all bars exceed the 95% confidence boundaries. The Durbin-Watson test (Table 2), which tests for first-order autocorrelation in the residuals, yields a statistic of 1.5262 ($p < 2.2 \times 10^{-16}$), far below the ideal value of 2.0. Therefore, the test rejects H_0 and concludes that autocorrelation is present in the residuals. Both results indicate that the errors are dependent, which violates assumption 4 and suggests inefficient parameter estimates and biased standard errors.

Table 2: Durbin-Watson Test for Autocorrelation

H_0	No autocorrelation is present in the residuals.	
H_1	True autocorrelation > 0 .	
DW Statistic	p -value	Significance
1.5262	$< 2.2 \times 10^{-16}$	***
Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05		

4.3.5 Assumption 5 - Zero Conditional Mean

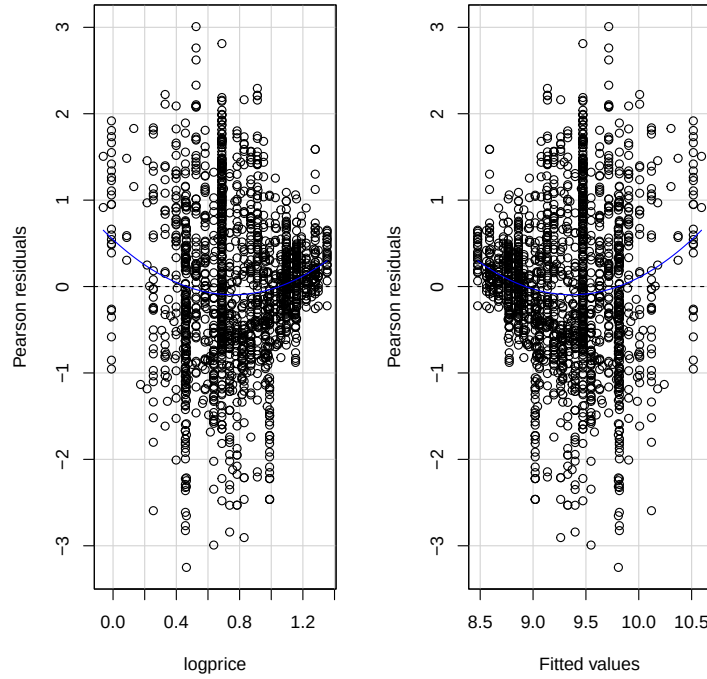


Figure 7: Residual Plots for Logprice and fitted values.

The Pearson Residual-Logprice plot in Figure 7 exhibits a U-shaped pattern, with residuals systematically positive at low and high values of Logprice and negative at intermediate values, indicating endogeneity. Endogeneity is likely present due to Logprice being correlated to omitted variables such as Brand. Therefore, $E[\varepsilon_i | X_i] \neq 0$, violating Assumption 5 and rendering the OLS estimator biased and inconsistent.

4.4 Model Results

Interpreting the results of model 1 shown in Table 16 (Appendix B.1) provide the following estimated model:

$$\widehat{\text{Logmove}} = 10.50110 - 1.49635 \cdot \text{Logprice} + \epsilon \quad (2)$$

The Intercept ($\beta_0 = 10.50110$) represents the predicted log-units sold when the actual price is \$1 ($\log(1) = 0$), serving as a baseline for demand at a standard price point. The coefficient for Logprice ($\beta_1 = -1.49635$) suggests that a 1% increase in price is associated with an approximately 1.50% decrease in units sold.

Table 3: Test for Significance of Logprice

H_0	$\beta_1 = 0$ (no relationship between Logprice and Logmove)	
H_1	$\beta_1 \neq 0$ (a relationship between Logprice and Logmove exists)	
Test Statistic (t)	p -value	Significance
-19.93	$< 2 \times 10^{-16}$	***

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

The p -value for Logprice (Table 3) is highly significant at the 5% level; therefore, H_0 is rejected, suggesting a highly significant relationship between Logprice and Logmove. However, as the OLS assumptions were violated by the presence of endogeneity, heteroscedasticity, and autocorrelation, the standard errors are biased, and the OLS estimates are inconsistent. Consequently, while the p -value suggests statistical significance, the test statistics are unreliable. Therefore, while the direction of β_1 is economically intuitive, its magnitude should be interpreted cautiously

The goodness of fit of the model is communicated by its R^2 values. The multiple R^2 of 0.1871 means that approximately 18.71% of the variation in Logmove is explained by Logprice. The Adjusted R^2 has a very similar value of 0.1866 as only one predictor was included. The low values for R^2 indicate that over 80% of the variation in units sold is **influenced** by factors other than price, such as brand identity, store differences, or promotional features.

5 Question 5a

Categorical variables with n levels are encoded as $n - 1$ binary dummy variables to satisfy the numerical input requirement of linear regression while avoiding perfect multicollinearity. Including all n levels as dummy variables creates perfect multicollinearity with the model's intercept, as their sum would equal 1 for every observation, making the model unable to produce unique coefficient estimates. By including $n - 1$ variables, one category is omitted and serves as the reference group and is absorbed into the intercept. This avoids multicollinearity as no dummy variable can be written as an exact linear combination of the others and the intercept.

Figure 8: Dummy Variable Coding for Brand

Brand	Dummy Variables	
	Minute Maid	Tropicana
Dominick's (ref.)	0	0
Minute Maid	1	0
Tropicana	0	1

Figure 9: Dummy Variable Coding for Store

Store	Dummy Variables			
	5	8	9	12
2 (ref.)	0	0	0	0
5	1	0	0	0
8	0	1	0	0
9	0	0	1	0
12	0	0	0	1

For the dummy encoding for Brand (Figure 8), we have 2 dummy variables for Minute Maid and Tropicana, with Dominick's selected as the reference category. Similarly, in Figure 9, 4 dummy variables are included for Stores 5, 8, 9 and 12, with Store 2 omitted as the reference category. Consequently, the intercept of the regression represents the expected value of the dependent variable for Dominick's and Store 2, establishing a baseline. The coefficients of the dummy variables measure the difference in the dependent variable relative to this baseline. A positive coefficient indicates higher expected sales compared to the baseline, while a negative coefficient indicates lower expected sales.

6 Question 5b

The OJ dataset has been randomly partitioned into a training set containing 75% of the observations (1,296) and a validation set containing 25% (432). The training set is used to estimate the regression model parameters, while the validation set is used to evaluate the model's performance. Training and testing on the same data risks overfitting, which occurs when the model captures noise rather than the true relationship between the predictor and response variables. By withholding 25% of the data, we can evaluate a model's generalisation to unseen observations and objectively compare model on independent validation data.

7 Question 5c

7.0.1 Model Specification

A multiple linear regression is estimated on the training set to evaluate the joint impact of price, feature, brand, store and week. The categorical variables are included via dummy variables.

$$\begin{aligned}\text{Logmove} = & \beta_0 + \beta_1 \text{Logprice} + \beta_2 \text{Feature}_1 + \beta_3 \text{Week} \\ & + \beta_4 \text{Brand}_{\text{MinuteMaid}} + \beta_5 \text{Brand}_{\text{Tropicana}} \\ & + \beta_6 \text{Store}_5 + \beta_7 \text{Store}_8 + \beta_8 \text{Store}_9 + \beta_9 \text{Store}_{12} + \varepsilon\end{aligned}\tag{3}$$

7.1 OLS Assumptions

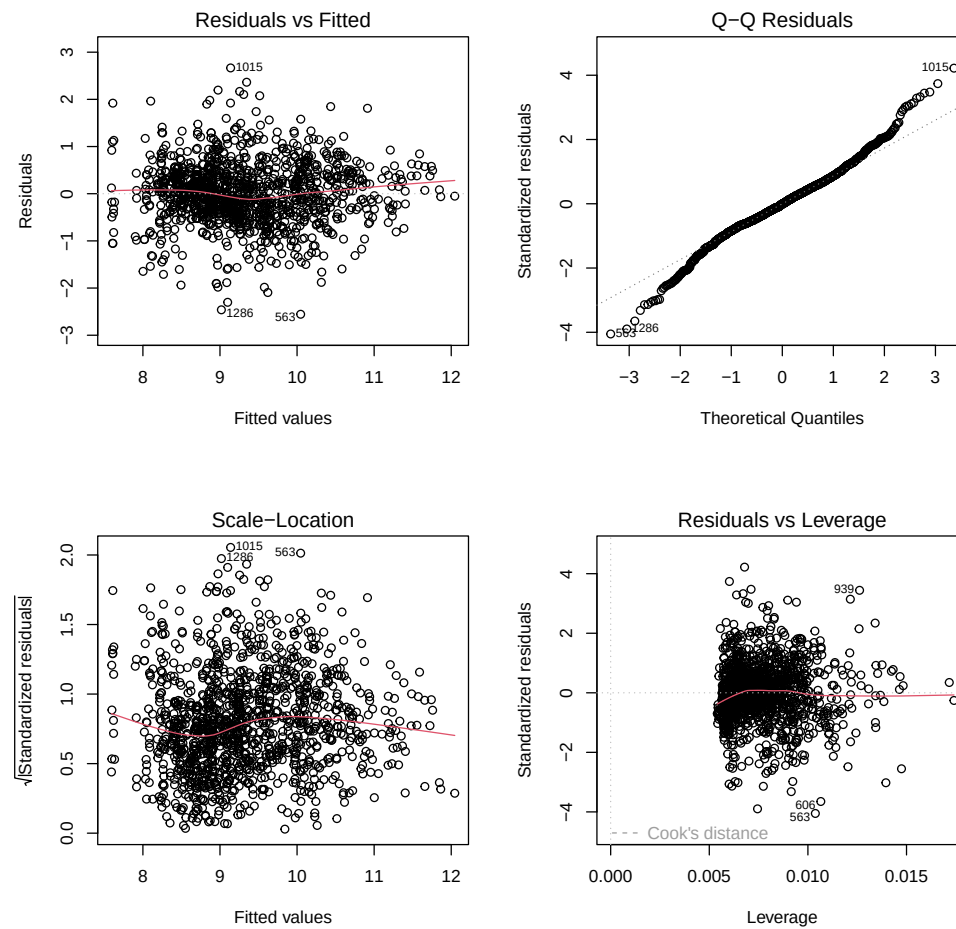
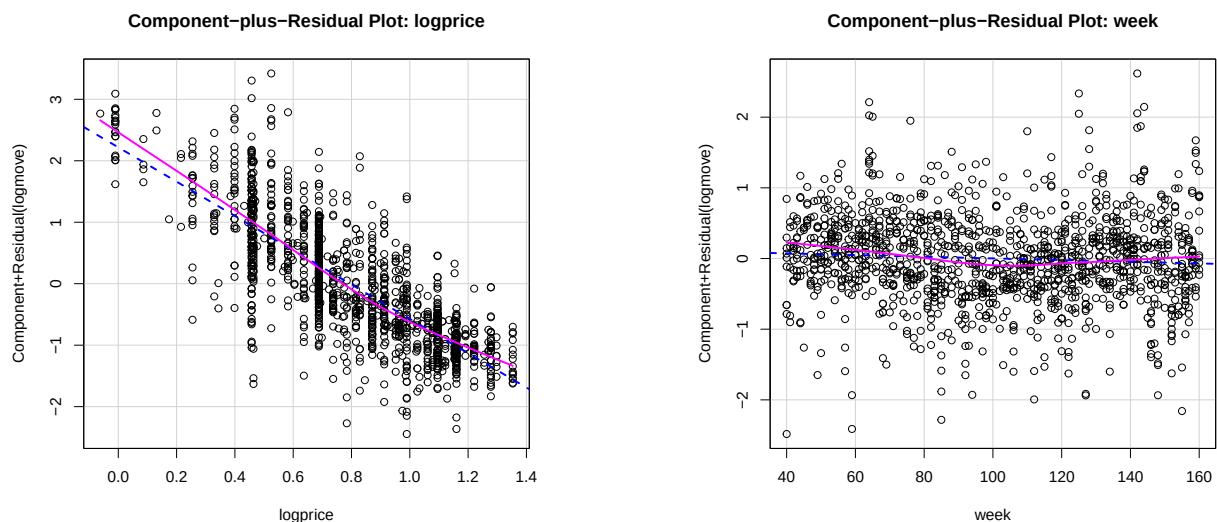


Figure 10: OLS Diagnostic Plots for the Multiple Linear Regression Model

7.1.1 Assumption 1: Linearity



(a) Component-plus-Residual Plot for Logprice

(b) Component-plus-Residual Plot for Week

Figure 11: Component-plus-Residual Plots examining linearity for selected predictors.

The Residual vs Fitted plot in Figure 10 shows the LOWESS deviating from zero, suggesting non-linearity. The CPR plots in Figure 11 confirm this for Logprice and week as the LOESS lines (purple) exhibit curvature and deviate from the Least-Squares lines (blue), indicating that these predictors have a non-linear relationship with Logmove.

Table 4: Curvature Tests for Predictor Linearity

H_0	The relationship between each predictor and Logmove is linear.		
H_1	The predictor has a non-linear relationship with Logmove.		
Variable	t-statistic	p-value	Significance
Logprice	3.9688	7.623×10^{-5}	***
Week	3.5908	3.421×10^{-4}	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

The curvature tests (Table 4) reject H_0 , confirming non-linearity for both Logprice and week. The Tukey test (Table 5), assesses the overall functional form of the model by examining the relationship between residuals and the squared fitted values.

Table 5: Tukey Test for Overall Model Specification

H_0	The model is correctly specified as a linear combination of the predictors.	
H_1	The model is misspecified; a non-linear functional form is required.	
t-statistic	p-value	Significance
2.5186	0.0118	.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1		

The Tukey test rejects H_0 of correct model specification. Therefore, the linearity assumption is violated, suggesting that the model's fit could be improved by non-linear transformations of Logprice and week or the inclusion of interaction terms.

7.1.2 Assumption 2 - Normality of the Residuals

The Q-Q plot in Figure 10 shows the residuals deviating from normality at the tails; however, due to the large sample size, the central limit theorem ensures that the sampling distribution of the coefficient estimators remains approximately normal, meaning that statistical inference remains valid.

7.1.3 Assumption 3 – Homoscedasticity

The Scale-Location plot (Figure 10) shows that the residual variance slightly decreases with higher fitted values, indicating minor heteroscedasticity. This suggests that the standard errors may be slightly inefficient, and the p-values should be interpreted with some caution. However, as the violation appears minor, it is unlikely to materially affect inference.

7.1.4 Assumption 4 - Independence of the residual error terms

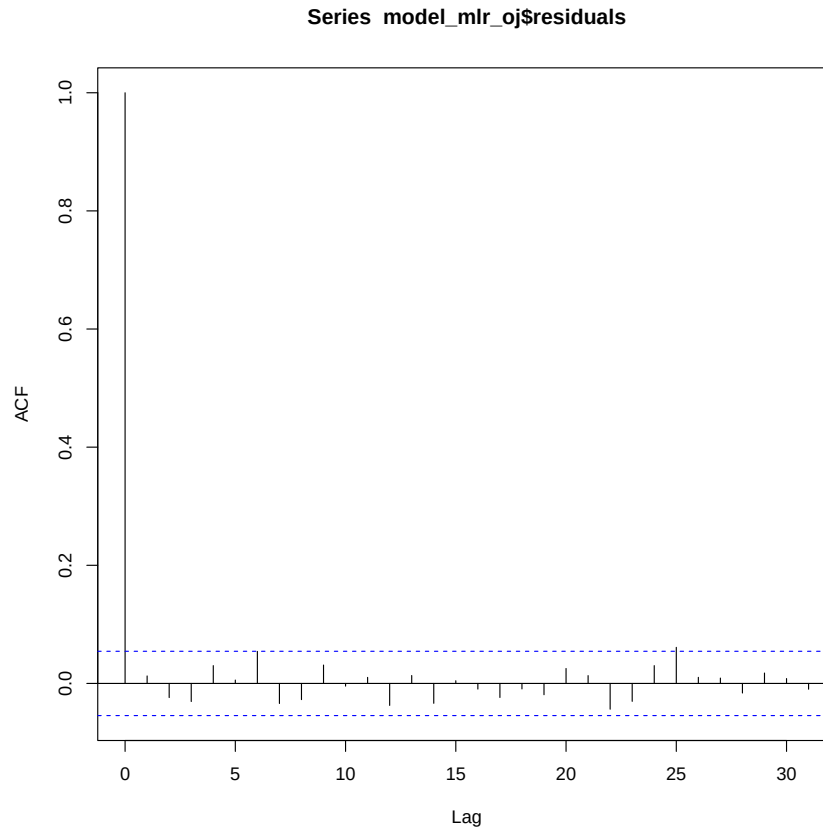


Figure 12: Autocorrelation Function (ACF) Plot of Residuals

Table 6: Durbin-Watson Test for Independence of Errors in Model 3

H_0	No first-order autocorrelation is present in the residuals.	
H_1	True autocorrelation > 0 .	
DW Statistic	p -value	Significance
1.974	0.3205	

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

The ACF plot (Figure 12) shows one bar (lag 25) marginally exceeding the 95% significance boundaries, however this is likely a false positive arising from chance. The Durbin-Watson test (Table 6), confirms this, with a test statistic of 1.974 ($p = 0.3205$) very close to the ideal value of 2.0, failing to reject H_0 of no autocorrelation. Taken together, these results confirm the absence of significant autocorrelation, satisfying assumption 4.

7.1.5 Assumption 5 - Zero Conditional Mean

Assumption 5 is satisfied, as in the residual vs fitted value plot (Figure 10), the LOWESS line remains close to 0, exhibiting negligible curvature, providing no strong evidence that $E[\epsilon_i|X_i] \neq 0$.

7.1.6 Assumption 6 - no (or little) multicollinearity

Table 7: Variance Inflation Factors (VIF) for Model 3

Variable	Df	GVIF ^{1/(2×Df)}
Logprice	1	1.579317
feature	1	1.075044
week	1	1.026163
brand	2	1.219385
store	4	1.005071

Multicollinearity is assessed using the adjusted Generalised Variance Inflation Factor (GVIF), which measures how much of the regression coefficients' variance is increased due to collinearity. Due to the presence of categorical variables, the adjusted GVIF (GVIF^{1/(2×Df)}) is used as it accounts for differing degrees of freedom (DF). The results in Table 7, show that all the values are well below the threshold of 2.24 (Appendix C). Store, Week and Feature show almost no multicollinearity (1.005-1.075). Brand (1.219) and Logprice (1.579) show only mild multicollinearity. Therefore, Assumption 6 is satisfied, and the standard errors and p-values for individual predictors are reliable and independently interpretable.

7.2 Model Results

The results of Model 3 (Table 17 in Appendix B.2), produce the following estimated model:

$$\begin{aligned}\widehat{\text{Logmove}} = & 10.4617 - 2.7912 \text{Logprice} + 0.8734 \text{feature}_1 - 0.0012 \text{week} \\ & + 0.8645 \text{Brand}_{\text{MinuteMaid}} + 1.6109 \text{Brand}_{\text{Tropicana}} \\ & + 0.0532 \text{Store}_5 + 0.3087 \text{Store}_8 \\ & - 0.0793 \text{Store}_9 + 0.4784 \text{Store}_{12}\end{aligned}\tag{4}$$

The intercept (β_0) is 10.4617. Since $\text{week} = 0$ is outside the observed range, the interpretation of the intercept is limited. The coefficient of Logprice ($\beta_1 = -2.7912, p < 2 \times 10^{-16}$) implies that a 1% increase in price reduces sales by 2.7912%, when holding all other variables constant, consistent with the economic laws of demand. The temporal trend, captured by week ($\beta_3 = -0.0012, p = 0.0233$), is statistically significant, suggesting an approximately $0.12\%(e^{-0.0012} - 1)$ decrease in sales per week. However, as Logprice and week were found to be non-linear, their coefficients should be treated as approximations of their true effects, with the actual relationships likely being more complex than a constant effect.

The presence of a promotional feature ($\beta_2 = 0.8734, p < 2 \times 10^{-16}$) is a highly significant driver of sales, corresponding to a roughly 139.50% increase in the number of units sold. By examining store-specific effects relative to Store 2, Store 8 ($\beta_7 = 0.3087, p = 5.18 \times 10^{-8}$) and Store 12 ($\beta_9 = 0.4784, p < 2 \times 10^{-16}$) exhibit significantly higher real sales, by 36.17% and 61.35% respectively. Conversely, sales at Stores 5 ($\beta_6 = 0.0532, p = 0.3491$) and 9 ($\beta_8 = -0.0793, p = 0.1630$) are not statistically different. Finally, Brand analysis reveals that Minute Maid ($\beta_4 = 0.8645, p < 2e - 16$) shows a significant 137.38% increase, and Tropicana ($\beta_5 = 1.6109, p < 2 \times 10^{-16}$) sees an approximately 400.73% increase in real sales when compared to Dominick's. Tropicana's large brand coefficient is particularly notable, as it also has the highest mean price, \$2.89 (Table 14, Appendix A.1), suggesting that its brand equity independently sustains sales volumes without reliance on price competition or promotional activity.

The adjusted R^2 (0.6086) means that approximately 60.86% of the variance in Logmove is explained by this model. The highly significant F-statistic ($F = 224.7, p < 2. \times 10^{-16}$) confirms that the explanatory variables are jointly significant. Although diagnostic tests found violations of linearity and homoscedasticity, the model provides a statistically and economically meaningful explanation of orange juice sales behaviour.

8 Question 5d

A quadratic term (Logprice^2) is added to Model 3 to test whether the non-linear effect of Logprice is significant. A quadratic term is appropriate here as it captures the U-shaped relationship seen in Figure 11 by allowing the marginal effect of Logprice on Logmove to vary with price.

$$\begin{aligned} \text{Logmove} = & \beta_0 + \beta_1 \text{Logprice} + \beta_2 \text{Logprice}^2 + \beta_3 \text{Feature} + \beta_4 \text{Week} \\ & + \beta_5 \text{MinuteMaid} + \beta_6 \text{Tropicana} \\ & + \beta_7 \text{Store5} + \beta_8 \text{Store8} + \beta_9 \text{Store9} + \beta_{10} \text{Store12} + \varepsilon \end{aligned} \quad (5)$$

The results from this model are found in Table 18 (Appendix B.3), with key coefficients summarised in equation 6:

$$\text{Logmove} = \dots - 3.8643 \text{Logprice} + 0.7452 \text{Logprice}^2 + \dots \quad (6)$$

Logprice^2 ($p = 7.62 \times 10^{-5}$) is highly significant, indicating that price elasticity is not constant, with the rate at which sales decline as prices rise varying across price levels. The positive coefficient indicates that demand becomes less price sensitive at higher prices as it partially offsets the negative coefficient of Logprice. Interestingly, the inclusion of Logprice^2 leads week lose its statistical significance ($p = 0.178$ compared to $p = 0.0233$ in Model 3). This suggests that weeks apparent significance in Model 3 may have been due to it capturing the non-linear price effect rather than a temporal trend. The adjusted R^2 increased from 0.6086 in model 3 to 0.6130, suggesting a marginally better model fit.

To justify the more complex non-linear model over the linear specification, an ANOVA test was performed (Table 8), evaluating if the inclusion of Logprice^2 in the complex model (Model 5) improves upon a nested model (Model 3) by analysing the reduction in the residual sum of squares (RSS).

Table 8: ANOVA Comparison Between Linear and Quadratic Models

H_0 : Logprice^2 does not significantly improve the model's fit (Model 3 is preferred).							
H_1 : Logprice^2 significantly improves the model's fit (Model 5 is preferred).							
Model	Res. Df	RSS	Df	Sum of Sq	F value	Pr(>F)	Significance
1: Linear	1286	516.81					
2: Quadratic	1285	510.55	1	6.2583	15.752	7.623×10^{-5}	***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1							

The test rejects H_0 , with the RSS reducing from 516.81 to 510.55, confirming that the quadratic model provides a statistically superior fit and that the non-linear effect of Logprice is significant.

9 Question 5e

An interaction term between Logprice and Brand is included to determine whether price elasticity differs across brands, given their varying price strategies.

$$\begin{aligned} \text{Logmove} = & \beta_0 + \beta_1 \text{Logprice} + \beta_2 \text{Feature} + \beta_3 \text{Week} \\ & + \beta_4 \text{MinuteMaid} + \beta_5 \text{Tropicana} \\ & + \beta_6 (\text{Logprice} \times \text{brand}_{\text{MinuteMaid}}) + \beta_7 (\text{Logprice} \times \text{brand}_{\text{Tropicana}}) \\ & + \beta_8 \text{Store5} + \beta_9 \text{Store8} + \beta_{10} \text{Store9} + \beta_{11} \text{Store12} + \varepsilon \end{aligned} \quad (7)$$

The interaction coefficients capture the difference in price elasticity between each brand and Dominick's (baseline), with the total price effect of Minute Maid and Tropicana being $\beta_1 + \beta_6$ and $\beta_1 + \beta_7$, respectively. The brand coefficients, β_4 and β_5 , represent the baseline sales differences when compared to Dominick's when $\text{Logprice} = 0$. A full regression output is provided in Table 19 (Appendix B.4), with key coefficients summarised in Equation 8:

$$\begin{aligned} \text{Logmove} = & \dots - 3.1062 \text{ Logprice} + \dots \\ & + 0.3470 (\text{Logprice} \times \text{Brand}_{\text{MinuteMaid}}) \\ & + 0.6977 (\text{Logprice} \times \text{Brand}_{\text{Tropicana}}) \\ & + 0.6703 \text{ MinuteMaid} + 1.0460 \text{ Tropicana} + \dots \end{aligned} \quad (8)$$

Logprice's coefficient (-3.1062 , $p < 2 \times 10^{-16}$), indicates that for Dominick's, there is a 3.11% decrease in units sold per 1% increase in price. The Minute Maid interaction term (0.3470 , $p = 0.116$) is not significant, providing no evidence that Minute Maid's price elasticity differs from Dominick's despite their different price points. This suggests that Minute Maid may compete primarily with promotions, such as features, instead of price positioning. Conversely, Tropicana's interaction term (0.6977 , $p = 5.78 \times 10^{-4}$) is statistically significant. Therefore, Tropicana is less price sensitive than Dominick's, as Tropicana's price elasticity is -2.41% ($-3.1062 + 0.6977 = -2.4085$). This could suggest that Tropicana has a loyal customer base who are willing to pay a premium, aligning with the finding from section 1 that Tropicana has a comparable sales volume to Dominick's despite a mean price that is nearly \$1.15 higher. Finally, Minute Maid (0.6703 , $p = 5.31 \times 10^{-5}$) and Tropicana (1.0460 , $p = 5.84 \times 10^{-9}$) both have significantly higher baseline sales than Dominick's when the price is held at \$1.

Table 9: ANOVA Comparison Between Linear and Interaction Models

H_0	The interaction terms do not significantly improve the model's fit.					
H_1	The interaction terms significantly improve the model's fit.					
Model	Res. Df	RSS	Df	Sum of Sq	F	Pr(>F)
1: Linear	1286	516.81				
2: Interaction	1284	512.05	2	4.7579	5.9653	0.002638 **
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1						

The ANOVA test (Table 9) rejects H_0 , concluding that the interaction terms significantly improve the model's fit. Additionally, the adjusted R^2 (0.6116) is higher than in Model 3 (0.6086), which supports the inclusion of the interaction terms.

10 Question 5f

Forward stepwise selection was performed on the training data. Starting from the null model, variables were added iteratively based on which addition results in the lowest AIC. AIC balances model fit against complexity by penalising additional parameters, with a lower AIC indicating a preferable model.

Table 10: Forward Stepwise Selection Based on AIC

Step	Added Variable	RSS	AIC	Model
0	-	1329.51	35.09	Null model: $\text{Logmove} \sim 1$
1	feature	934.23	-420.20	$\text{Logmove} \sim \text{feature}$
2	Logprice	826.90	-576.36	$\text{Logmove} \sim \text{feature} + \text{Logprice}$
3	brand	575.47	-1042.16	$\text{Logmove} \sim \text{feature} + \text{Logprice} + \text{brand}$
4	store	518.88	-1168.31	$\text{Logmove} \sim \text{feature} + \text{Logprice} + \text{brand} + \text{store}$
5	week	516.81	-1171.50	$\text{Logmove} \sim \text{feature} + \text{Logprice} + \text{brand} + \text{store} + \text{week}$

All five variables were selected (Table 10), validating Model 3's specification. The order in which variables are added is economically informative. Feature was added first, producing the largest reduction in AIC (35.09 to -420.20), indicating that features are the most dominant predictor of sales. Logprice was added next, followed by brand, reflecting the significance of price sensitivity and brand identity. Despite some insignificant dummies, store was added fourth, capturing location-specific demand differences with a collective AIC reduction of 126.15. Finally, week was added with a negligible reduction in AIC (-1168.31 to -1171.50), which suggests no meaningful temporal trend beyond what is already captured by the other variables. This is consistent with week's marginal significance in Model 3.

11 Question 5g

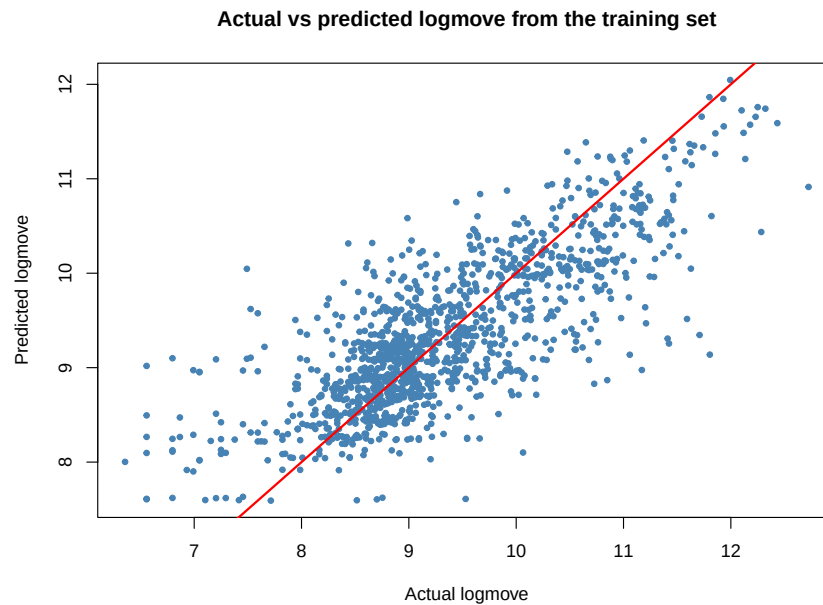


Figure 13: Actual vs. Predicted Logmove for the training dataset. The red line represents perfect prediction (Predicted = Actual).

Model 3 achieved an MSE of 0.3988, corresponding to a typical prediction error of approximately 0.6315 log units ($\sqrt{0.3988}$). This MSE could be optimistic, as the model was trained and tested on the same data. Figure 13 shows the points cluster reasonably well along the perfect prediction line, confirming that the model captures the general trend in sales. Accuracy is highest for average Logmove values (8.5–9.5), as the errors in this region are minimal. At extreme values (Logmove < 8 or > 10), the increased spread of points indicates weaker predictive power, suggesting the influence of omitted factors.

12 Question 5h

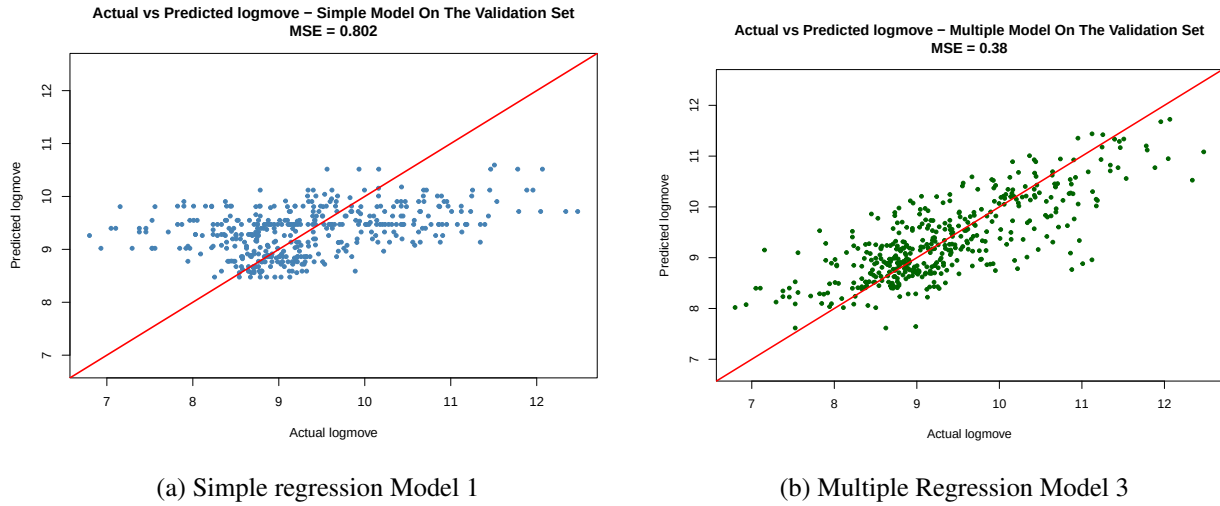


Figure 14: The red line represents perfect prediction ($y = x$).

Table 11: Model Performance Comparison: Training vs Validation Set

Model	Train MSE	Val MSE	Train RMSE	Val RMSE
Model 1	0.8286	0.8028	0.9103	0.8960
Model 3	0.3988	0.3800	0.6315	0.6165

Both models were trained on the training set and evaluated on the disjoint and independent validation set. Model 1 achieved an MSE of 0.8028 (RMSE = 0.8960), whereas Model 3 achieved an MSE of 0.3800 (RMSE = 0.6165), a 52.6% reduction in MSE, with Model 3's predictions on average 0.2795 log units closer to the actual values. Additionally, Table 11 shows practically no difference between each model's training and validation MSEs, strongly suggesting that neither model is overfitting to the data.

Figure 14b plots the predicted Logmove values from Model 3 against the actual Logmove values. The points cluster tightly around the perfect prediction line, demonstrating a high level of accuracy. However, this model does encounter some issues with over and underestimating Logmove at extreme values, evidenced by the increased errors at the tails. In contrast, Figure 14a shows the corresponding plot for Model 1. The predicted values exhibit much greater deviation from the perfect prediction line, displaying horizontal banding because the model relies solely on Logprice to explain demand variation, leading it to predict similar demand levels across observations with differing characteristics. Model 1's failure to capture high and low demand scenarios is indicative of omitted variable bias, as by ignoring the impact of brand-specific pricing and promotional features, the model fails to account for the primary drivers of sales volatility for orange juice.

To conclude, Model 3 should be used, as it offers superior predictive accuracy and generalises well to unseen data, with the inclusion of additional explanatory variables such as feature, brand, store and week significantly improving the model's ability to account for the heterogeneity in sales behaviour.

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A Descriptive Statistics

A.1 Question 1: OJ Dataset Summary Statistics Tables

Table 12: Statistical Summary of variables in the Orange Juice Dataset (OJ)

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Week	40.00	70.75	101.00	100.76	131.00	160.00
Logmove	6.356	8.691	9.101	9.316	9.952	12.724
Price	0.940	1.790	2.170	2.300	2.810	3.870
Variables	Counts / Frequencies					
Brand	Dominick's: 576, Minute Maid: 576, Tropicana: 576					
Feature	0 (No): 1321, 1 (Yes): 407					
Store	Store 8: 354, Store 9: 351, Store 5: 348, Store 12: 345, Store 2: 330					

Table 13: Summary Statistics of variables in the Orange Juice Dataset (OJ) by Store

Metric	Store 12	Store 9	Store 8	Store 5	Store 2
Logmove					
Mean	9.516	9.178	9.571	9.176	9.131
Max	12.477	12.338	12.724	11.742	11.855
Pricing (USD)					
Mean Price	2.413	2.240	2.183	2.312	2.359
Max Price	3.870	3.390	3.190	3.660	3.870
Features					
Feature Incidence Rate	23.5%	23.6%	23.7%	23.3%	23.6%

Table 14: Summary Statistics of variables in the Orange Juice Dataset (OJ) by Brand

Metric	Dominick's	Minute Maid	Tropicana
Pricing (USD)			
Mean Price	1.746	2.260	2.894
Min Price	0.940	1.290	1.490
Max Price	2.690	3.170	3.870
Sales & Marketing			
Mean Logmove	9.236	9.374	9.339
Feature (Count)	148	165	94

A.2 Question 3: Store 2 Brand Analysis Table

Table 15: Summary Statistics of Logmove by Brand (Store 2)

Brand	Min.	1st Qu. (Q_1)	Median	Mean ($\bar{x}$)	3rd Qu. (Q_3)	Max.	IQR	Std. Dev. (σ)
Dominick's	6.557	8.486	9.122	9.074	9.667	11.855	1.181	1.071
Minute Maid	7.897	8.407	8.891	9.091	9.605	11.820	1.198	0.848
Tropicana	8.166	8.744	9.023	9.227	9.676	10.850	0.932	0.646

B Model Results

B.1 Question 4: Model Results Table

Table 16: Simple Linear Regression Results: Logmove vs. Logprice

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.50110	0.06332	165.83	$< 2e - 16$ ***
Logprice	-1.49635	0.07508	-19.93	$< 2e - 16$ ***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
Metric	Value			
Multiple R^2	0.1871			
Adjusted R^2	0.1866			
F-statistic	$F(1, 1726) = 397.2, p < 2.2 \times 10^{-16}$			
Observations	1728			
Residual Std. Error	0.9072 on 1726 DF			

B.2 Question 5c: Model Results Table

Table 17: Multiple Linear Regression Results for Logmove

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.4617	0.1007	103.880	$< 2e-16$ ***
logprice	-2.7912	0.0966	-28.907	$< 2e-16$ ***
feature1	0.8734	0.0445	19.638	$< 2e-16$ ***
week	-0.0012	0.0005	-2.272	0.0233 *
brandminute.maid	0.8645	0.0508	17.029	$< 2e-16$ ***
brandtropicana	1.6109	0.0641	25.124	$< 2e-16$ ***
store5	0.0532	0.0568	0.937	0.3491
store8	0.3087	0.0564	5.478	$5.18e-08$ ***
store9	-0.0793	0.0568	-1.396	0.1630
store12	0.4784	0.0566	8.460	$< 2e-16$ ***
<i>Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1</i>				
Fit Statistics		Values		
Observations		1296		
Residual Std. Error		0.6339 on 1286 DF		
Multiple R^2		0.6113		
Adjusted R^2		0.6086		
F-statistic		224.7 ($p < 2.2e-16$)		

B.3 Question 5d: Model Results Table

Table 18: Regression Results for Quadratic Model of Logmove

Variable	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	10.7229	0.1198	89.483	< 2e-16	***
logprice	-3.8643	0.2869	-13.467	< 2e-16	***
I(logprice ²)	0.7452	0.1878	3.969	7.62e-05	***
feature1	0.8785	0.0442	19.858	< 2e-16	***
week	-0.0007	0.0005	-1.347	0.178	
brandminute.maid	0.8994	0.0512	17.554	< 2e-16	***
brandtropicana	1.5688	0.0646	24.275	< 2e-16	***
store5	0.0636	0.0565	1.126	0.260	
store8	0.3338	0.0564	5.919	4.15e-09	***
store9	-0.0525	0.0569	-0.924	0.356	
store12	0.4781	0.0562	8.503	< 2e-16	***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					
Residual Std. Error: 0.6303 on 1285 DF					
Multiple R-squared: 0.6160, Adjusted R-squared: 0.6130					
F-statistic: 206.1 on 10 and 1285 DF, p-value < 2.2e-16					

B.4 Question 5e: Model Results Table

Table 19: Regression Results for Logprice–Brand Interaction Model

Variable	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	10.6202	0.1127	94.243	< 2e–16	***
logprice	-3.1062	0.1398	-22.221	< 2e–16	***
brandminute.maid	0.6703	0.1653	4.055	5.31e-05	***
brandtropicana	1.0460	0.1785	5.861	5.84e-09	***
feature1	0.8792	0.0444	19.809	< 2e–16	***
week	-0.0011	0.0005	-2.114	0.0347	*
store5	0.0494	0.0566	0.874	0.383	
store8	0.3084	0.0562	5.488	4.90e-08	***
store9	-0.0763	0.0566	-1.348	0.178	
store12	0.4770	0.0563	8.468	< 2e–16	***
logprice:brandminute.maid	0.3470	0.2205	1.574	0.116	
logprice:brandtropicana	0.6977	0.2022	3.451	0.000578	***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					
Residual Std. Error: 0.6315 on 1284 DF					
Multiple R-squared: 0.6149, Adjusted R-squared: 0.6116					
F-statistic: 186.3 on 11 and 1284 DF, p-value < 2.2e–16					

C GVIF^{1/(2×Df)} Threshold explanation

For the adjusted GVIF^{1/(2×Df)}, the following thresholds are applied:

GVIF^{1/(2×Df)} < 2.24: Implies no problematic multicollinearity.

GVIF^{1/(2×Df)} ≥ 2.24: Implies Problematic multicollinearity present.

The threshold of 2.24 corresponds to a standard VIF of 5 (since $\sqrt{5} \approx 2.24$).

D R Code

Libraries and Data Loading

```
1
2 library(AER)
3 library(dplyr)
4 library(lmtest)
5 library(car)
6 library(ggplot2)
7 library(GGally)
8 library(MASS)
9 install.packages("MLmetrics")
10 library(MLmetrics)
11
12 #Loading the dataset
13
14 setwd("C:/Users/adamr/OneDrive - University of Edinburgh/Year 3/Predictive
    Analytics/PAB-Coursework1")
15
16 oj <- read.csv("C:/Users/adamr/OneDrive - University of Edinburgh/Year 3//
    Predictive Analytics/PAB-Coursework1/oj.csv")
17
18 #Incase plots break after clearing plot output
19 graphics.off()
20
21 head(oj)
22
23 oj$brand <- as.factor(oj$brand)
24 oj$store <- as.factor(oj$store)
25 oj$feature <- as.factor(oj$feature)
```

Question 1

```
1
2 dim(oj)
3 #summary of entire dataset
4 summary(oj)
5 #summary of each brand
6 summary(oj[o$brand == "dominicks", ])
7 summary(oj[o$brand == "minute.maid", ])
8 summary(oj[o$brand == "tropicana", ])
9 #summary for each store
10 summary(oj[o$store == 12, ])
11 summary(oj[o$store == 9, ])
12 summary(oj[o$store == 8, ])
13 summary(oj[o$store == 5, ])
14 summary(oj[o$store == 2, ])
```

Question 2

```
1 pdf("Q2_Histogram_Logmove.pdf", width = 8, height = 6)
2 h <- hist(oj$logmove,
3           col = "steelblue",
4           main = "Histogram of Logmove (Log of number of units sold)",
5           xlab = "Log of Unit Sales",
6           breaks = 12)
7 # Generate x values
8 x_vals <- seq(min(oj$logmove), max(oj$logmove), length = 100)
9 # Add properly scaled normal curve for dist comparison
10 lines(x_vals,
11        dnorm(x_vals,
12              mean = mean(oj$logmove),
13              sd = sd(oj$logmove)) *
14              length(oj$logmove) *
15              diff(h$breaks)[1],
16              col = "red",
17              lwd = 2)
18 dev.off()
19
20 # histogram of raw sales data in order to compare skewedness
21 pdf("Q2_Histogram_Logmove_rawsales(optional).pdf", width = 8, height = 6)
22 hist(exp(oj$logmove),
23       col = "darkred",
24       main = "Histogram of Units Sold (Original Scale)",
25       xlab = "Actual Units Sold",
26       breaks = 30)
27 dev.off()
```

Question 3

```
1 store2_data <- subset(oj, store ==2)
2
3 tapply(store2_data$logmove, store2_data$brand, summary)
4 tapply(store2_data$logmove, store2_data$brand, sd)
5
6 pdf("Q3_Logmove_by_Brand_Store2.pdf", width = 8, height = 6)
7 plot(as.factor(store2_data$brand), store2_data$logmove,
8      main = "Logmove by Brand for Store 2",
9      xlab = "Brand",
10     ylab = "Logmove",
11     col = c("Orange", "red", "yellow"))
12 dev.off()
13
14 #the histogram to investigate skewness and support box median line claim
15 summary(store2_data) # for finding optimal bin number, using same rule as in
    Q2
16
17 #Dominick's Histogram
18 pdf("Hist_Dominicks_Store2.pdf", width = 5, height = 4)
19 dom_store2_data <- subset(store2_data, brand == "dominicks")
20 hist(dom_store2_data$logmove,
21      main = "Dominick's Logmove Distribution",
22      xlab = "Logmove",
23      col = "orange",
24      breaks = 8)
25 dev.off()
26
27 #Minute Maid Histogram
28 pdf("Hist_MinuteMaid_Store2.pdf", width = 5, height = 4)
29 mm_store2_data <- subset(store2_data, brand == "minute.maid")
30 hist(mm_store2_data$logmove,
31      main = "Minute Maid Logmove Distribution",
32      xlab = "Logmove",
33      col = "red",
34      breaks = 8)
35 dev.off()
36
37 #Tropicana Histogram
38 pdf("Hist_Tropicana_Store2.pdf", width = 5, height = 4)
39 trop_store2_data <- subset(store2_data, brand == "tropicana")
40 hist(trop_store2_data$logmove,
41      main = "Tropicana Logmove Distribution",
42      xlab = "Logmove",
```

```
43     col = "yellow",  
44     breaks = 8)  
45 dev.off()
```

Question 4

```
1
2 oj$logprice <- log(oj$price) #Creating Logprice variable
3
4 model_simple <- lm( logmove ~ logprice, data = oj)
5
6 #Checking assumptions:
7 pdf("Q4_Simple_Model_Diagnostics.pdf", width = 8, height = 8)
8 par(mfrow=c(2,2))
9 plot(model_simple)
10 dev.off()
11
12 par(mfrow=c(1,1))
13
14 #Supporting linearity plot
15
16 pdf("Q4_Simple_Model_Linearity.pdf", width = 8, height = 8)
17 ggplot(oj, aes(x = logprice, y = logmove)) +
18   geom_point(alpha = 1, color = "black") +
19   geom_smooth(method = "lm", color = "red", se = FALSE) + # Linear line
20   geom_smooth(method = "loess", color = "blue", se = FALSE) +
21   labs(title = "Linearity Check: Logmove vs Logprice",
22        subtitle = "Red = Linear Model, Blue = Loess Smoother")
23 dev.off()
24
25 #checking for autocorrelation
26 dwtest(model_simple)
27
28 #Visually checking autocorrelation
29 pdf("Q4_Simple_Model_ACF.pdf")
30
31 acf(model_simple$residuals)
32
33 dev.off()
34
35 # Zero Conditional mean:
36 pdf("Q4_Simple_Model_Residual_Plots.pdf")
37 residualPlots(model_simple) # also produces the Tukey and Curvature tests
38 dev.off()
39
40 summary(model_simple)
```

Question 5a

```
1
2 #as.factor() is a data structure used to categorize data, R automatically
   creates N-1 dummy variable used earlier in code
3 # for categorical variables with N levels
4
5 # Check the levels of the Brand variable
6 levels(oj$brand)
7
8 levels(oj$store)
9 # View the dummy coding matrix for Brand
10 contrasts(oj$brand)
11 contrasts(oj$store)
12
13 numberOfBrands <- length(unique(oj$brand))
14 numberOfStores <- length(unique(oj$store))
15
16 #k-1 req
17 dum_brand <- numberOfBrands - 1
18 dum_store <- numberOfStores - 1
19
20 print(paste(("Number of Brands"),numberOfBrands, "Number of Dummy Variables
   for Brands", dum_brand))
21 print(paste(("Number of Stores"),numberOfStores, "Number of Dummy Variables
   for Stores", dum_store))
```

Question 5b

```
1
2 set.seed(123) # set to seed to ensure i get the same results every time
3
4 train_ind <- sample.int(n = nrow(oj), size = floor(.75*nrow(oj)), replace = F)
   #sample takes samples randomly according to seed, replace F makes sure
   they're disjoint
5
6 #creating the training and validation set
7 oj_train = oj[train_ind, ]
8 oj_test = oj[-train_ind, ] #contains all entries not in the training set
9
10 #checking dimensions
11 dim(oj)
12 dim(oj_train)
13 dim(oj_test)
```

Question 5c

```
1
2 model_mlr_oj <- lm(logmove ~ logprice + feature + week + brand + store, data=
   oj_train)
3
4 #Checking Ols assumptions
5
6 #plotting model with diagnostic plots
7 levels(oj_train$feature)
8
9 pdf("Q5c_Multiple_Model_Diagnostics.pdf", width = 8, height = 8)
10
11 par(mfrow=c(2,2))
12 plot(model_mlr_oj)
13
14 dev.off()
15
16 par(mfrow=c(1,1))
17 crPlots(model_mlr_oj)
18
19 #investigating linearity in the continuous variables
20
21 # Component-plus-residual plot for logprice
22 pdf("Q5c_CRPlot_logprice.pdf", width = 6, height = 6)
23 crPlot(model_mlr_oj, "logprice", main = "Component-plus-Residual Plot:
   logprice")
24 dev.off()
25
26 # Component-plus-residual plot for week
27 pdf("Q5c_CRPlot_week.pdf", width = 6, height = 6)
28 crPlot(model_mlr_oj, "week", main = "Component-plus-Residual Plot: week")
29 dev.off()
30
31 #Checking Independence of Errors ( autocorrelation)
32
33 dwtest(model_mlr_oj)
34
35 pdf("Q5c_Multiple_Model_ACF.pdf", width = 8, height = 8)
36 acf(model_mlr_oj$residuals)
37 dev.off()
38
39 #Checking for MultiColinearity
40
41 #using VIF
```

```

42 vif(model_mlr_oj)
43 #Visually
44
45 x <- (oj_train[,-4])
46
47 pdf("Q5c_Multiple_Model_ggpairs.pdf", width = 8, height = 8)
48
49 ggpairs(x)
50
51 dev.off()
52
53 #Checking for Zero Conditional Mean of Errors assumption
54 pdf("Q5c_Multiple_Model_ResidualPlots.pdf", width = 8, height = 8)
55
56 residualPlots(model_mlr_oj) # also produces the Tukey and Curvature tests
57
58 dev.off()
59
60 #model results
61
62 summary(model_mlr_oj)

```

Question 5d

```
1
2 model_mlr_oj_n1 <- lm(logmove ~ logprice + I(logprice^2) + feature + week +
3   brand + store, data= oj_train)
4
5 summary(model_mlr_oj_n1)
6
7 # Anova test to compare quadratic model to with the original model
8 anova( model_mlr_oj,model_mlr_oj_n1)
```

Question 5e

```
1
2 model_mlr_oj_it <- lm(logmove ~ logprice * brand + feature + week + store,
3   data= oj_train)
4
5 summary(model_mlr_oj_it)
6
7 anova(model_mlr_oj, model_mlr_oj_it)
```

Question 5f

```
1
2 model_null <- lm(logmove ~1, data=oj_train) #defining the starting null model
3
4 model_full <- lm(logmove ~ logprice + feature + week + brand + store, data =
5   oj_train)
6
7 #performing the stepwise AIC
8
9 step_forward <- stepAIC(model_null,
10   scope = list(lower = model_null, upper = model_full),
11   direction = "forward")
12
13 summary(step_forward)
```

Question 5g

```
1
2 pred_train = predict(model_mlr_oj, oj_train)
3
4 mse_train <- MSE(y_pred = pred_train, y_true = oj_train$logmove)
5 print(paste("Training MSE:", mse_train))
6
7 pdf("Q5g_Actual_vs_Predicted_Training.pdf", width = 8, height = 6)
8
9 plot_actual_predicted <- plot(x =oj_train$logmove,
10                               y= pred_train,
11                               main ="Actual vs predicted logmove from the
12                                   training set",
13                               xlab="Actual logmove",
14                               ylab="Predicted logmove",
15                               pch=20,
16                               col = "steelblue")
17 # reference line
18 abline(0, 1, col = "red", lwd = 2)
19
20 dev.off()
```

Question 5h

```
1
2 #retraining the simple model on the Testing set
3 model_simple_train <- lm( logmove ~ logprice, data = oj_train)
4
5 #Accuracy for Simple Model
6 pred_simple <- predict(model_simple_train, oj_test)
7 mse_simple <- MSE(y_pred = pred_simple, y_true = oj_test$logmove)
8
9 #Accuracy for Multiple Model, validation MSE
10 pred_multiple <- predict(model_mlr_oj, oj_test)
11 mse_multiple <- MSE(y_pred = pred_multiple, y_true = oj_test$logmove)
12
13 # Compute training MSEs
14 pred_simple_train <- predict(model_simple_train, oj_train)
15 pred_multiple_train <- predict(model_mlr_oj, oj_train)
16
17 mse_simple_train <- MSE(y_pred = pred_simple_train, y_true = oj_train$logmove)
18 mse_multiple_train <- MSE(y_pred = pred_multiple_train, y_true = oj_train$
    logmove)
19
20
21
22 #building table for results
23 model_comparison <- data.frame(
24   Model = c("Simple", "Multiple"),
25   Train_MSE = round(c(mse_simple_train, mse_multiple_train), 4),
26   Val_MSE = round(c(mse_simple, mse_multiple), 4),
27   Train_RMSE = round(c(sqrt(mse_simple_train), sqrt(mse_multiple_train)), 4),
28   Val_RMSE = round(c(sqrt(mse_simple), sqrt(mse_multiple)), 4)
29 )
30
31 #comparison results table
32 print(model_comparison)
33
34 #setting y scale to be the same for each for better side by side comparison
35
36 y_min <- min(c(pred_simple, pred_multiple, oj_test$logmove))
37 y_max <- max(c(pred_simple, pred_multiple, oj_test$logmove))
38
39 #plots
40 pdf("Q5g_Actual_vs_Predicted_Validation_Simple.pdf", width = 8, height = 6)
41 plot(x = oj_test$logmove,
42       y = pred_simple,
```

```

43     ylim = c(y_min, y_max),
44     main = paste("Actual vs Predicted logmove - Simple Model On The
        Validation Set\nMSE =", round(mse_simple, 3)),
45     xlab = "Actual logmove",
46     ylab = "Predicted logmove",
47     pch = 20,
48     col = "steelblue")
49 # Reference line
50 abline(0, 1, col = "red", lwd = 2)
51 dev.off()
52
53 # Plot actual vs predicted for Multiple Model
54 pdf("Q5g_Actual_vs_Predicted_Validation_Multiple.pdf", width = 8, height = 6)
55 plot(x = oj_test$logmove,
56      y = pred_multiple,
57      ylim = c(y_min, y_max),
58      main = paste("Actual vs Predicted logmove - Multiple Model On The
        Validation Set\nMSE =", round(mse_multiple, 3)),
59      xlab = "Actual logmove",
60      ylab = "Predicted logmove",
61      pch = 20,
62      col = "darkgreen")
63 # Reference line
64 abline(0, 1, col = "red", lwd = 2)
65 dev.off()

```